MIL-L-21514C (Wep)
2 August 1963
Superseding
MIL-L-21514B (Wep)
4 April 1961

MILITARY SPECIFICATION

LAUNCHER TUBE, MISSILE, MARK 4, MODS 0, 1, 2, AND 3 AND MARK 8 MODS 0 AND 1

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification states the requirements for the launcher tubes which form a part of the Launching System, Missile, Mark 17 Mods 0, 1, and 2 and Mark 21 Mods 0 and 2. (See 6.1.)

2. APPLICABLE DOCUMENTS

2.1 Documents - The following documents of the issue in effect on date of invitations for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-S-21516	Support Ring for Launcher Tube, Missile
MIL-B-21518	Bellows Assembly, Flexible
MIL-C-21529	Cable Assemblies, Electrical

FSC 1440

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2.1 (cont)

NAVY DEPARTMENT

BUREAU OF NAVAL WEAPONS, Code Ident 10001

OS 10087	Detonator Box Assembly
BUREAU OF SHIPS	General Specifications for Ships of the United States Navy, Section S9-1, Welding and Allied Processes

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

DRAWINGS

NAVY DEPARTMENT

BUREAU OF NAVAL WEAPONS, Code Ident 10001

LD 419542	Launcher Tube, Missile, Mark 4 Mod 0
LD 419543	Launcher Tube, Missile, Mark 4 Mod 1
LD 540870	Launcher Tube, Missile, Mark 4 Mod 2
LD 540871	Launcher Tube, Missile, Mark 4 Mod 3
LD 540950	Launcher Tube, Missile, Mark 8 Mod 0

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2.1 (Cont)

LD 540951	Launcher Tube, Missile Mark 8 Mod 1
LD 540954	Retract Mechanism, Main Umbilical Assembly for Mark 3 Mod 0 Missile
LD 540991	Retract Mechanism, Tactical Umbilical Installation for Mk 1 & 2 Missiles
LD 541006	Base Umbilical Bridge Installation
LD 543802	Base Umbilical Bridge Installation
Dwg 2025360	Simulated Missile for Umbilical Retract Test
Dwg 2025384	Hydro Test Fixture Entire Tube
SA-2440479	Vibration Test Fixture
SA-2586697	Drop Shock Test Fixture

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2.1 (Cont)

PUBLICATIONS

NAVY DEPARTMENT

BUREAU OF NAVAL WEAPONS

NAVWEPS OP 400	General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment
NAVWEPS OP 2230	Workmanship and Design Practices for Electronic Equipment
NAVORDINST 4355.1	Equipment for Use by Government Inspectors in Final Inspection of Naval Ordnance Items
BUWEPSINST 4355.5	Bureau of Naval Weapons Calibration Program
Standards Laboratory Information Manual	

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other Publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

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3. REQUIREMENTS

3.1 Description - The launcher tube is a steel cylinder 30 feet 11-1/8 inches long with a bore of 57 inches, formed by joining five tube sections, and includes access door assemblies, an umbilical retract mechanism, as specified (see 6.2), a base umbilical bridge installation where required and specified, (see 6.2), a support ring assembly (MIL-S-21516), flexible bellows assemblies (MIL-B-21518), interlock switches, electrical cable assemblies (MIL-C-21529) and on the Mark 4 tube, a detonator box assembly (OS 10087). The launcher tube supports the missile during stowage, has openings to permit missile servicing, and guides the missile during launching.

3.2 General Requirements

3.2.1 Preproduction Samples - Unless otherwise specified in the contract or order, preproduction samples of the launcher tube shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification (refer to 4.2.1). At the discretion of the procuring agency, the requirement for a preproduction sample may be waived and shall then be satisfied by substitution of a production unit chosen at random by the Government Inspector.

3.2.2 Materials - Unless otherwise specified in the contract, all materials and purchased parts entering into the final assembly of the launcher tube shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BuWeps LD 419542, 419543, 540870, 540871, 540950, or 540951, LD 540991 or LD 540954 and 541006 or LD 540954 and 543802.

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3.2.3 Finishes, Coatings, Construction, and Dimensions - Finishes, coatings, construction, and dimensions for the launcher tube and associated parts shall be in accordance with the applicable drawings, specifications, and publications listed in BuWeps LD 419543, 419543, 540870, 540871, 540950 or 540951, LD 540991, or LD 540954 and 541006 or LD 540954 and 543802.

3.2.4 Marking - Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in BuWeps LD 419542, 419543, 540870, 540871, 540950 or 540951, LD 540991, or LD 540954 and 541006 or LD 540954 and 543802.

3.2.5 Interchangeability - Individual assemblies and piece parts of assemblies fabricated in agreement with their applicable drawings and specifications shall be interchangeable with like numbered parts.

3.2.6 Gages - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, requisition, or order, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.7 Threads - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Request for Deviations - The procedures for requesting deviations from this specification or any specification, publication, material, or process specified herein shall be as set forth in NAVWEPS OP 400.

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3.2.9 Conflicting Requirements - In event of conflict between this specification and any specification, publication, or drawing referenced herein, this specification shall govern.

3.2.10 Tolerances - Unless otherwise specifically noted, tolerances and accuracies stated herein are absolute.

3.3 Performance Requirements

3.3.1 Deleted

3.3.2 Hydrostatic Pressure - The launcher tube, including the umbilical retract mechanism, intertube seals, flexible bellows assembly, support ring assembly, and pressure pickup assemblies, and with all doors locked shut, shall withstand an internal proof pressure of 195 to 205 psig for 10 minutes and shall show no leakage or permanent deformation.

3.3.3 Dielectric Strength - After installation on the launcher tube, the door-assembly interlock switches, the electrical cable assemblies (MIL-C-21529), the detonator box assembly on Mark 4 tubes (Code Ident 10001 OS 10087), and the electrical components of the support ring assembly (MIL-S-21516) shall withstand a test voltage of 325-volts AC (rms) for 1 minute and shall show no damage.

3.3.4 Insulation Resistance - After installation on the launcher tube, the door-assembly interlock switches, the electrical cable assemblies (MIL-C-21529), the detonator box assembly on Mark 4 tubes (Code Ident 10001 OS 10087), and the electrical components of the support ring assembly (MIL-S-21516) shall have an insulation resistance of not less than 3 megohms, referred to normal operating temperatures, when subjected to a test voltage of 500-volts DC.

3.3.5 Access Door Assemblies

3.3.5.1 Mechanical Operation - The door assembly shall operate from locked shut to fully open, or from fully open to locked shut, within 40 seconds and shall not require a force exceeding 50 pounds at any time during the cycle.

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requirements

3.3.5.2 Interlock Switches - The door-assembly interlock switches shall initiate an electrical signal when the clamping ring is rotated counterclockwise so that the clamping ring lugs engage the access-door lugs by at least 2 inches (chordal measurement) and, when the clamping ring is rotated clockwise, shall discontinue the signal before lug engagement reaches 1-1/2 inches (chordal measurement). The switches shall not give false signals during launch tube pressurization or during the launching stroke.

3.3.5.3 Operating Life - The door assembly shall have an operating life of 3,000 cycles, when tested from locked shut to fully open to locked shut, without maintenance other than lubrication of the mechanical moving parts. However, replacement of the O-ring is permitted prior to the leakage test. No sign of leakage or excessive wear shall be permitted for compliance with this requirement.

3.3.6 Umbilical Retract Mechanisms

3.3.6.1 Motion - The umbilical retract mechanism shall move the disconnected umbilical away from the missile in a manner which prevents the retract mechanism or the umbilical from contacting the moving missile. Thereafter the retract mechanism shall show no damage and shall be capable of meeting all performance requirements of this specification.

3.3.6.2 Load - The umbilical retract mechanism shall not impose a load on the moving missile exceeding 600 pounds in the direction parallel to the longitudinal axis of the missile.

3.3.6.3 Operating Life - The umbilical retract mechanism shall withstand 100 cycles of operation with no maintenance other than lubrication.

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3.3.7 Base Umbilical Bridge Installation - The base umbilical bridge installation mounted in the normal operating position shall withstand a vertically upward force of 300 pounds applied at the center of the bridge for 1 second; shall withstand a vertically downward force of 3,000 pounds applied at the connector retaining bolts for 30 seconds, shall show no damage and thereafter shall be capable of meeting all requirements of this specification.

3.4 Environmental Conditions

3.4.1 Vibration, Operating Condition

3.4.1.1 Launcher Tube - The launcher tube including the umbilical retract mechanism shall withstand vibration in accordance with Standard MIL-STD-167, within the range of 5 to 25 cycles-per-second, shall show no damage and thereafter shall be capable of meeting all performance requirements of this specification.

3.4.1.2 Base Umbilical Bridge Installation - The base umbilical bridge installation shall withstand vibration in accordance with Standard MIL-STD-167, within the range of 5 to 25 cycles-per-second, shall show no damage and thereafter shall be capable of meeting all performance requirements of this specification.

3.4.2 Vibration, Shipping Condition

VIBRATION CRITERIA (MAX) FOR PACKAGING

Type of Transportation	Vertical		Longitudinal		Lateral	
	cps	g	cps	g	cps	g
Truck	200	2.5	150	1.5	200	1.0
	10	1.0	10	1.0	10	1.0
Rail	60	2.0	60	1.0	30	1.0
	10	1.0	10	0.5	10	0.5
Aircraft	60-100	1.0	60-100	1.0	60	1.0
	100-200	0.5			100	0.5
Ship (surface)	10	0.5	Negligible		10	0.25
	30	1.5			30	0.75

Each vibration condition shall be maintained for 10 minutes.

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3.4.2.1 Launcher Tube - The launcher tube including the umbilical retract mechanism, packaged and packed in accordance with section 5 herein, shall withstand the vibrations specified in vibration criteria, shall show no damage, and thereafter shall be capable of meeting all performance requirements of this specification.

3.4.2.2 Base Umbilical Bridge Installation - The base umbilical bridge installation, packaged and packed in accordance with section 5 herein, shall withstand the vibrations specified in vibration criteria, shall show no damage, and thereafter shall be capable of meeting all performance requirements of this specification.

3.4.3 Shock, Operating Condition

3.4.3.1 Launcher Tube - The launcher tube (with missile installed), complete with the umbilical retract mechanism, shall withstand the equivalent steady-state acceleration of 5g downward and 3g upward (including gravity) axially and 4g transversely. This requirement shall be considered satisfied after performance of the test specified in 4.6.3.1 followed by meeting the performance requirements specified in 3.3.3 through 3.3.5.2 and 3.3.6.1.

3.4.3.2 Base Umbilical Bridge Installation - The base umbilical bridge installation shall withstand acceleration of 5g downward and 3g upward (including gravity) with a rise time of 7 to 10 milliseconds, an athwartships acceleration of 4g with a rise time of 15 ± 5 milliseconds, and a fore and aft acceleration of 2g with a rise time of 25 ± 5 milliseconds. The base umbilical bridge installation shall show no damage and thereafter shall be capable of meeting all performance requirements of this specification.

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3.4.4 Shock, Shipping Condition

TRANSPORTATION SHOCK CRITERIA FOR PACKAGING

Type of Transportation	Acceleration g (max) Direction of Transportation Parallel	Lateral	Vertical
Rail	20	2	5 (Gravity included)
Aircraft or Truck	5	1	3 (Gravity included)

Handling sling loadings shall be 3g vertical and 2g lateral, maximum.

3.4.4.1 Launcher Tube - The launcher tube assembly including the umbilical retract mechanism, packaged and packed as specified in section 5 herein, shall withstand the shock specified in shock criteria, shall show no damage, and thereafter shall meet all performance requirements of this specification.

3.4.4.2 Base Umbilical Bridge Installation - The base umbilical bridge installation, packaged and packed as specified in section 5 herein, shall withstand the shock specified in shock criteria, shall show no damage, and thereafter shall be capable of meeting all performance requirements of this specification.

3.4.5 Inclination

3.4.5.1 Access Door Assemblies - The launcher tube access-door assemblies shall meet the requirements of 3.3.5.1, and the hinge-locking mechanism shall prevent door movement while inclined 10 degrees from the vertical in any direction.

3.4.5.2 Umbilical Retract Mechanism - The umbilical retract mechanism shall operate as specified in 3.3.6.1 and 3.3.6.2 while subjected to inclination up to 10 degrees from the vertical in any direction and shall show no damage.

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3.4.6 Corrosive Atmosphere

3.4.6.1 Launcher Tube - The launcher tube including the umbilical retract mechanism shall withstand the salt spray test, Specification MIL-E-5272, Procedure I, and thereafter shall be capable of meeting all performance requirements of this specification.

3.4.6.2 Base Umbilical Bridge Installation - The base umbilical bridge installation shall withstand the conditions specified in 3.4.6.1 and thereafter shall show no corrosion detrimental to operation.

3.4.7 Heat/Pressure Cycle

3.4.7.1 Launcher Tube - The launcher tube, Mark 8 Mods 0 and 1, shall withstand the environment of 5 consecutive heat/pressure cycles without damage, and thereafter shall be capable of meeting all performance requirements of this specification. A heat/pressure cycle shall consist of a change in pressure from atmospheric pressure to 115 ± 15 psig and return to atmospheric coincident with a temperature change from ambient to 325 ± 25 degrees Fahrenheit to 210 ± 20 degrees Fahrenheit within 1 second, and then to ambient temperature within 10 minutes.

3.4.7.2 Base Umbilical Bridge Installation - The base umbilical bridge installation (LD 542802) shall withstand 5 consecutive heat/pressure cycles as specified in 3.4.7.1 and thereafter shall be capable of meeting all performance requirements of this specification.

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3.4.8 Submersion

3.4.8.1 Launcher Tube - The areas of the launcher tube subjected to flooding shall withstand submersion in sea water at temperatures within the range between +28 and +86 degrees Fahrenheit for 90 consecutive days and thereafter shall be capable of meeting all performance requirements of this specification.

3.4.8.2 Base Umbilical Bridge Installation - The base umbilical bridge installation shall withstand the conditions specified in 3.4.8.1 and thereafter shall be capable of meeting all performance requirements of this specification.

3.5 Cleanup - Prior to and after assembly, all parts, components, and the assembly shall be thoroughly cleaned of metal chips and other foreign matter. Burrs and sharp edges shall be removed.

3.6 Workmanship - The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with this specification. Particular attention shall be paid to neatness and thoroughness of marking of parts and assemblies, painting, machine-screw assemblies, welding, brazing, and freedom of parts from burrs and sharp edges. Welding and brazing shall comply with Section S9-1 of General Specifications for Ships of the United States Navy. NAVWEPS OP 2230 shall be used as a guide for determining acceptable workmanship of electrical and electronic parts.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's Inspection - Unless otherwise specified herein, the contractor is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Government Inspection - The Government Inspector will make such inspections as are necessary to determine that components and assemblies are in accordance with the requirements of the contract, pertinent drawings, and specifications. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in NAVWEPS OP 400. Production tests of final assemblies shall be on a 100-percent basis. No sampling is permitted for acceptance inspection purposes. Only those assemblies which pass final inspection shall be accepted by the Navy; a rejected assembly may be reworked and submitted in accordance with NAVWEPS OP 400. Failed parts and components shall be submitted to the Material Review Board for dispostion.

4.2.1 Preproduction Sample - One preproduction sample, specified in 3.2.1 and 4.6, shall be submitted to the Government Inspector for preproduction tests at an activity designated in the contract, or by the procuring activity, for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order. Disposition of the test sample shall be at the discretion of the Material Review Board.

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4.3 Classification of Tests - The inspection and testing of the launcher tube shall be classified as follows:

4.3.1 Production Tests - These tests are to be accomplished on each launcher tube being submitted for acceptance under the contract. Production tests shall be either performed by the manufacturer and witnessed by the Government Inspector, or these tests may be conducted by the Government Inspector. These tests are detailed in 4.5.

4.3.2 Preproduction Tests - Preproduction tests are those which are accomplished on the sample selected as specified in 4.2.1 and 4.6 as representative of the production of the launcher tube to determine that the production meets the requirements of this specification. These tests are detailed in 4.5 and 4.6. Acceptance shall be based on no defects in the sample. Failure of the sample to comply with these requirements will result in the rejection of the lot or the cessation of production until determination of cause of failure is made and appropriate correction is approved by the procuring activity.

4.3.3 Tests After Delivery - Unless otherwise specified in the contract, the Government reserves the right to conduct any or all of the tests specified herein as deemed necessary after installation of the launcher tube in the submarine. Any defects discovered before, during, or after these tests, due to the contractor's failure to comply with drawings or specifications for this equipment, shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

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4.4 Special Test Equipment - The following items of special test equipment are required to perform the acceptance tests set forth in this specification:

BuWeps Dwg 2025360	Simulated Missile for Umbilical Retract Test
BuWeps Dwg 2025384	Hydro Test Fixture Entire Tube
SA-2440479	Vibration Test Fixture
SA-2586697	Drop Shock Test Fixture

4.5 Production Tests

4.5.1 Responsibility - The contractor or Government Inspector shall conduct production tests as specified in 4.3.1 to assure that the launcher tube is in compliance with the requirements of 3.3.2 through 3.3.5.2, and 3.3.6.1.

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4.5.2 Test Conditions - Unless otherwise specified, the launcher tube, the umbilical retract mechanism, and the base umbilical bridge installation shall be subjected to production tests under the following conditions:

Temperature:	Room ambient, +65 to +85 degrees Fahrenheit
Altitude:	Normal ground atmospheric pressure
Vibration:	None
Humidity:	Relative humidity of 50 percent or less at +78 degrees Fahrenheit

4.5.3 Test and Inspection Equipment and Facilities - The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all production tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified herein, shall be of sufficient accuracy and quality to permit performance of the required production tests.

4.5.3.1 Calibration - Measuring equipment and instrumentation shall be calibrated as required by NAVORDINST 4355.1, BUWEPSINST 4355.5, and in accordance with the policies specified in the Standards Laboratory Information Manual issued by the Bureau of Naval Weapons Representative, Pomona, California.

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4.5.4 Test Procedures

4.5.4.1 Deleted

4.5.4.2 Hydrostatic Tests - The following tests (at 1-1/2 times working pressure) shall be performed to meet the requirements specified in 3.3.2.

4.5.4.2.1 Assembled Tube - Each assembled tube shall be tested by installing the hydro test fixture, BuWeps Dwg 2025384, in the assembled tube to test each joint and penetration by applying a hydrostatic proof pressure of 195 to 205 psig for 10 minutes. The tube shall show no leakage or permanent deformation.

4.5.4.2.2 Deleted

4.5.4.3 Dielectric Strength Tests - After installation on the launcher tube, each interlock switch, electrical cable assembly, the detonator box assembly on Mark 4 tubes, and support ring assembly shall be tested to meet the requirements specified in 3.3.3 by applying a test voltage of 325-volts AC (rms) between the terminals and a ground for 1 minute.

4.5.4.4 Insulation Resistance Tests - After installation on the launcher tube, each interlock switch, electrical cable assembly, the detonator box assembly on Mark 4 tubes, and support ring assembly shall be tested to meet the requirements specified in 3.3.4 by applying a test voltage of 500-volts DC between the electrical terminals and a ground.

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4.5.4.5 Access Door Assemblies

4.5.4.5.1 Operating Force Test - Each access-door assembly installed on a tube shall be tested to meet the requirements specified in 3.3.5.1 by moving the door from locked shut to locked open or locked open to locked shut within 40 seconds and with a force not greater than 50 pounds at any time in the cycle.

4.5.4.5.2 Calibration Test - Each access-door assembly installed on the tube shall be tested to meet the requirements specified in 3.3.5.2 by connecting the interlock switch to suitable test lights and manually rotating the door clamping ring. The door-closed test light shall come ON when the clamping ring is rotated counterclockwise so that the clamping ring lugs engage the access-door lugs by at least 2 inches (chordal measurement). When the clamping ring is rotated clockwise, the door-open test light shall come ON before lug engagement reaches 1-1/2 inches (chordal measurement).

4.5.4.6 Operational Motion Test - Umbilical Retract Mechanism - To meet the requirements of 3.3.6.1, each umbilical retract mechanism shall be installed on a tube fourth section and operated with the use of test fixture BuWeps Dwg 2025360.

4.6 Preproduction Tests - Tests on the tube fourth section shall be deemed adequate to demonstrate quality of the assembled tube. The detonator box assembly, when part of the tube assembly, is normally mounted on the third tube section. During tests it shall be mounted on the fourth tube section in a manner simulating the actual mounting.

4.6.1 Vibration Tests - Operating Condition

4.6.1.1 Launcher Tube - To meet the requirements specified in 3.4.1.1, one tube fourth section complete with guidance access-door assembly, umbilical retract mechanism, umbilical cover plate, and on Mark 4 tubes the detonator box assembly, shall be subjected to vibration tests as specified in Standard MIL-STD-167 and shall show no damage.

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4.6.1.2 Base Umbilical Bridge Installation - One base umbilical bridge installation shall be subjected to vibration tests to meet the requirements of 3.4.1.2.

4.6.2 Vibration, Shipping Condition

4.6.2.1 Launcher Tube - To meet the requirements of paragraph 3.4.2.1, an engineering analysis of the packaging and packing of the launcher tube may be submitted to the procuring agency for approval and in substitution of the test.

4.6.2.2 Base Umbilical Bridge Installation - To meet the requirements of 3.4.2.2, an engineering analysis of the packaging and packing of the launcher tube may be submitted to the procuring activity for approval and in substitution of the test.

4.6.3 Shock Tests - Operating Condition

4.6.3.1 Launcher Tube - To meet the requirements specified in 3.4.3.1, one tube fourth section complete with guidance access-door assembly, umbilical retract mechanism, umbilical coverplate, and on Mark 4 tubes the detonator box assembly, shall be assembled with the test fixtures SA-2440479 and SA-2586697 and tested as follows:

The tube section, in the horizontal position and with the access door open, shall be swung so that the bottom rim of the tube section strikes a solid stop, producing a longitudinal deceleration of 5 to 5-1/2g. The tube section shall be reversed and with the access door open shall be swung so that the top rim of the tube section strikes the solid stop, producing a longitudinal deceleration of 3 to 3-1/2g. The tube section, in the normal position and with the access door open, shall be swung against a solid stop to produce a deceleration of 4 to 4-1/2g. Each of these tests shall then be repeated with the access door closed.

4.6.3.2 Base Umbilical Bridge Installation - One base umbilical bridge installation shall be subjected to shock tests to meet the requirements of 3.4.3.2.

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4.6.4 Shock, Shipping Condition

4.6.4.1 Launcher Tube - To meet the requirements of paragraph 3.4.4.1, an engineering analysis of the packaging and packing of the launcher tube may be submitted to the procuring agency for approval and in substitution of the test.

4.6.4.2 Base Umbilical Bridge Installation - To meet the requirements of 3.4.4.2, an engineering analysis of the packaging and packing of the bridge installation may be submitted to the procuring agency for approval and in substitution of the test.

4.6.5 Inclination Test - To meet the requirement specified in 3.4.5, one tube fourth section, complete with guidance access-door assembly, umbilical retract mechanism, and umbilical coverplate, shall be tilted so that the edge directly below the guidance access-door rests on the floor, and the opposite edge is approximately 12 inches above the floor. In this position, the access door, hinge-locking mechanism, and retract mechanism shall be tested for satisfactory operation and the test shall be repeated in each quadrant.

4.6.6 Corrosive Atmosphere Test

4.6.6.1 Launcher Tube - One launcher tube fourth section, complete with guidance access-door assembly, umbilical retract mechanism, and umbilical coverplate, shall be tested to meet the requirement specified in 3.4.6.1 by exposing both sides of the tube section to a salt atmosphere, in accordance with Specification MIL-E-5272, for 50 consecutive hours. Upon completion of the 50 hours, the retract mechanism and the access door operating hardware shall be disassembled, washed with fresh water and dried. The equipment shall show no corrosion detrimental to continued operation.

4.6.6.2 Base Umbilical Bridge Installation - One base umbilical bridge installation will be tested to meet the requirements of 3.4.6.2.

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4.6.7 Heat/Pressure Cycle Test

4.6.7.1 Launcher Tube - One launcher tube, or bottom tube assembly, shall be installed in a prototype launcher (Peashooter) or an acceptable equivalent test apparatus and shall be subjected to a heat/pressure test environment to meet the requirements of 3.4.7.1.

4.6.7.2 Base Umbilical Bridge Installation - One base umbilical bridge installation under the conditions specified in 4.6.7.1 shall be subjected to test environment to meet the requirements of 3.4.7.2.

4.6.8 Submersion Test

4.6.8.1 Launcher Tube - One tube fourth section, complete with guidance access-door assembly, umbilical retract mechanism, and umbilical coverplate shall be tested to meet the requirements specified in 3.4.8.1 by closing all openings and keeping the tube filled with sea water to the level of the center of the guidance-access-door at temperatures within the range between +28 and +86 degrees Fahrenheit for 90 consecutive days. Upon completion of the 90 days, the tube section shall be disassembled, washed with fresh water and dried. The equipment shall show no corrosion detrimental to continued operation.

4.6.8.2 Base Umbilical Bridge Installation - To meet the requirements of 3.4.8.2, one umbilical bridge installation shall be submerged in sea water at temperatures within the range between +28 and +86 degrees Fahrenheit for 90 consecutive days.

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the umbilical retract mechanism shall withstand the pull

4.6.9 Operational Load Test, Umbilical Retract Mechanism - To meet the requirements of 3.3.6.2, one umbilical retract mechanism shall be installed on a tube fourth section and operated with test fixture BuWeps Dwg 2025360.

4.6.10 Operation Test, Base Umbilical Bridge Installation - To meet the requirements of 3.3.7, one base umbilical bridge installation shall be subjected to an upward vertical load of 300 pounds for 1 second and to a downward vertical load of 3,000 pounds for 30 seconds.

4.6.11 Operating Life Test

4.6.11.1 Access Door Assemblies - To meet the requirements of 3.3.5.3, one access-door assembly shall be subjected to an operating test from locked shut to fully open to locked shut for 3,000 cycles without maintenance other than lubrication of the mechanical moving parts. After testing, the door shall meet all other requirements of paragraph 3.3.5.1 and 3.3.5.2.

4.6.11.2 Umbilical Retract Mechanism - One umbilical retract mechanism, installed on a tube fourth section and mounted on the simulated missile test stand (BuWeps Dwg 2025360), shall be tested to meet the requirements of 3.3.6.3 by operating through 100 cycles, with no maintenance other than lubrication, and shall meet the requirements of 3.3.6.1.

4.7 Rejection - Failure of the equipment to meet any of the requirements or tests of this specification shall be considered cause for rejection.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging - Preservation and packaging of the launcher tube shall be Level A or Level C as follows:

5.1.1 Level A - Level A shall be as specified by the procuring activity.

5.1.2 Level C - Cleaning, drying, preservation and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing

5.2.1 Level A - Level A shall be as specified by the procuring activity.

5.2.2 Level C - The launcher tube, preserved as specified in 5.1.2, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or other common-carrier regulations applicable to the mode of transportation.

5.3 Marking - Unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

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6. NOTES

6.1 Intended Use - To stow, maintain, and permit servicing of POLARIS Missiles aboard ship and to guide the missile during launching. The Missile Launcher Tube Mk 4 Mod 0 and 1 is used with Missile Launching System Mk 17 Mod 0; Mk 4 Mod 2 and 3 with Missile Launching System Mk 17 Mod 1; Mk 8 Mod 0 and 1 with Missile Launching System Mk 17 Mod 2 and Mk 21 Mod 0 and 2. Capabilities of the various marks and mods are as follows:

Launcher Tube	Drawing	Missile Launching Capability
Mk 4 Mod 0 w/stool	LD 419542	POLARIS Mk 2 Mod 0 (Tactical) Mk 1 Mod 0 (Tactical)
Mk 4 Mod 1 w/stool	LD 419543	POLARIS Mk 2 Mod 0 (Tactical) (Test and Evaluation) Mk 1 Mod 0 (Tactical) (Test and Evaluation
Mk 4 Mod 2 w/stool	LD 540870	POLARIS Mk 2 Mod 0 (Tactical) Mk 3 Mod 0 (Tactical) Mk 1 Mod 0 (Tactical
Mk 4 Mod 3 w/stool	LD 540871	POLARIS Mk 2 Mod 0 (Tactical) (Test and Evaluation) Mk 3 Mod 0 (Tactical) (Test and Evaluation Mk 1 Mod 0 (Tactical) (Test and Evaluation)
Mk 8 Mod 0	LD 540950	POLARIS Mk 2 Mod 0 (Tactical) Mk 3 Mod 0 (Tactical)
Mk 8 Mod 1	LD 540951	POLARIS Mk 2 Mod 0 (Tactical) (Test and Evaluation) Mk 3 Mod 0 (Tactical) (Test and Evaluation)

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6.2 Ordering Data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Mark and Mod of launcher tube.

(c) Umbilical retract mechanism and accessories to be furnished with the tube.

1. Retract Mechanism, Tactical Umbilical Installation for Missiles Mk 1 and 2 (LD 540991) should be specified for all tubes herein when the tube is to handle POLARIS Mk 1 or Mk 2 Missile.

2. Retract Mechanism, Main Umbilical Assembly for Mk 3 Mod 0 Missile (LD 540954) plus Base Umbilical Bridge Installation (LD 541006) should be specified for launcher tube Mk 4 Mod 2 and 3 when the tube is to handle POLARIS Mk 3 Mod 0 Missile.

3. Retract Mechanism, Main Umbilical Assembly for Mk 3 Mod 0 Missile (LD 540954) plus Base Umbilical Bridge Installation (LD 543802) should be specified for launcher tube Mk 8 Mod 0 and 1 when the tube is to handle POLARIS Mk 3 Mod 0 Missile.

(d) Exceptions to this specification, applicable drawings, and other documents.

(e) Selection of applicable levels of preservation, packaging, and packing.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 46-R308

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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BUREAU OF NAVAL WEAPONS
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deliver the fluids specified herein at the flow pressure components to